HW3 Demand Analysis

Problem 4.1 Elasticity.

A.) A price reduction will increase both units demanded & total revenue. (True)

Since price is elasticity is -5 demand is elastic The percentage increase in quantity demanded exceeds the percentage decrease in price. Both units demanded total revenue increase.

B) A 5% reduction in computer price will cause a 20% increase in software demand. (False)

The cross price elasticity of computer demand with respect to software price is -4.

$$E_{PC} = \frac{\% \Delta Q}{\% \Delta P} = -4$$

$$\% \Delta Q_{comp} = (-4)(-5\%) = +20\% \text{ increase}$$

computer demand.

c) Demand for personal computers is price elastic and computers are cyclical normal goods (True)

When income rises by 1% computer demand increased by approx. 2.5%. When income falls by 1% demand decreases by appoximately 2.5%

D) falling software prices will increase revenues received by sellers of both computers and software. (false

Because computers and software are complements, a reduction in software prices tends to increase demand for computers.

E) A 2% price reduction is necessary to overcome a 1% decline in income. (False)

$$\%\Delta Q_S = E_y \times \%\Delta Y$$

$$= 2.5 \times (1) = -2.5\%$$

$$\%\Delta Q_D = E_p \times \%\Delta P$$

$$2.5\% = (-5) \times \%\Delta P$$

$$\%\Delta P = -0.5\%$$

-0.5% price reduction not a 2% price reduction

Problem 4.4 Cross - Price elasticity.

Given	Value
B.B Jeans original price	$140
B.B Jeans initial sales	10,000 units

Competitors price ⇒ 137 $

Competitors price reduction ⇒ 52 $

$$P_C = 137 - 52 = \$85$$

Part A

$$E_{AB} = \frac{\% \Delta Q_A}{\% \Delta P_B}$$

product A is B.B Jean's garment bag.

product B is the competitors garment bag.

$$\% \Delta Q_A = \frac{4.800 - 10,000}{10,000} \times 100$$

$$= -52\%$$

$$\% \Delta P_B = \frac{85 - 137}{137} \times 100 = -37.96\%$$ (percentage change in competitors price)

Cross price elasticity

$$E_{AB} = \frac{-52\%}{-37.96\%} \approx +1.37$$

The positive cross price elasticity indicates that the two garment bags are substitutes. When competitor reduces its price customers switch away from B.B Jeans.

Part B 4.4

sales increase from 4,800 to 6,000 units

$$\% \Delta Q = \frac{6,000 - 4,800}{4,800} \times 100$$

$$= 25\ \%$$

price falls from $140 to $130

$$\% \Delta P = \frac{130 - 140}{140} \times 100 = -7.14\%$$

$$E_p = \frac{25\ \%}{-7.14\%} \approx -3.50$$

Demand price is elastic $|E_p| = 3.50 > 1$. A 1% decrease in price is associated with approximately a 3.5% increase in quantity demanded using the same percentage change method.

Part C

$$\% \Delta Q = \frac{10,000 - 6,000}{6,000} \times 100 = 66.67\%$$

$$-3.5 = \frac{66.67\ \%}{\% \Delta P}$$

$$\% \Delta P = \frac{66.67}{-3.5} = -19.05\%$$

$$P_{(new)} = 130\ (1 - 0.1905)$$

$$P_{new} \approx 105.24\ \$$$

Reduction = 130 - 105.24

= 24.76 $

Problem 4.6 Income Elasticity.

Part A

$$\%\Delta Q_d = \frac{50-30}{30} \times 100$$

$$\boxed{\%\Delta Q_d = 66.67\%}$$

$$\%\Delta Y = \frac{58.500 - 55,500}{55,500} \times 100$$

$$\boxed{\%\Delta Y = 5,4054\%}$$

income elasticity

$$E_y = \frac{66.67\%}{5.4054\%}$$

$$E_y \simeq 12.33$$

The implied income elasticity is approximately +12.33. This is extremely high: the director's forecast implies that a 1% increase in household income leads to about a 12.33% increase in carpet demanded.

Part B

$$\%\Delta Q_d = \frac{30-50}{50} \times 100$$

$$\%\Delta Q_d = -40\%$$

$$\%\cdot\Delta P = \frac{16.50 - 15.50}{15.50} \times 100$$

$$\boxed{\%\Delta P = 6.4516\%}$$

$$E_p = \frac{-40\%}{6.4516\%}$$

$$\boxed{E_p \simeq -6.20}$$

Demand is highly price elastic because $|E_p| = 6.20 > 1$. The implied elasticity suggests that a 1% price increase reduces quantity demanded by approximately 6.2%, using the initial-value method.

Part C

$$TR = P \times Q$$

$$TR = \$16.50 \times 30 \text{ million}$$

$$TR = \underline{\$495 \text{ million}}$$

Because demand is elastic, a further price increase would reduce total revenue, assuming the elasticity remains applicable and other factors stay constant.

Problem 4.8 Cross - Price Elasticity.

Price elasticity demand $E_p = \frac{\%\,\Delta Q \text{ appetizers}}{\%\,\Delta P \text{ appetizers.}}$

Cross - Price elasticity $E_{BA} = \frac{\%\,\Delta Q \text{ beverages}}{\%\,\Delta P \text{ appetizers.}}$

Part A

$$\%\,\Delta Q_A = \frac{150 - 90}{90} \times 100 \qquad \%\,\Delta Q_A = 66.67\%$$

$$\%\,\Delta P_A = \frac{10 - 12}{12} \times 100 = -16.67\%$$

$$E_p = \frac{66.67}{-16.67} = -4$$

Appetizer demand price is elastic because $|E_p| = 4 > 1$

Part B

$$\%\,\Delta Q_B = \frac{600 - 300}{300} \times 100.$$

$$\%\,\Delta Q_B = 100\% \qquad \%\,\Delta P = -16.67\%$$

$$E_{BA} = \frac{100\%}{-16.67\%} = -6.$$

increase in orders is proportionally larger than the price decrease.

Cross-price is negative meaning appetizers and beverages are complementary goods.

Parte 4.8.

$TR = Q \times P$

$TR_1 = 12 \times 90 = \$\ 1{,}080$ (before the original price)

$TR_2 = 10 \times 150 = 1500\ \$$

Appetizer has revenue increase from 1,080 to 1500 $ per day.

Estimate demand at $8

$$\%\Delta P = \frac{8-10}{10} \times 100 = -20\%$$

$Q_{new} = 150(1+0.80)$

$= 270$

$\%\Delta Q = (-4)(-20)\% = +80\%$

$TR_{new} = 8 \times 270 = \$\ 2{,}160$ per day.

Cross-price = $\%\Delta Q_B = (-6)(-20\%) = +120\%$

$Q_{B new} = 600(1+1.20) = 1320$

beverage sales are estimated to increase from 600 to 1320 units per day. Additional price reduction to $8 is expected to increase both appetizer and beverage.